# Introduction to Machine Learning

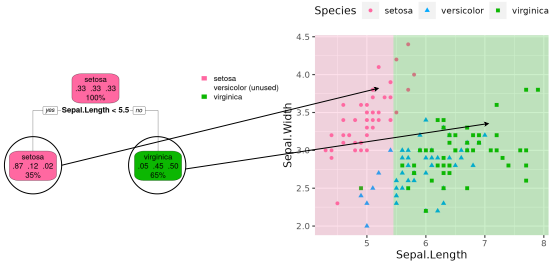
# CART: Growing a Tree

## Learning goals

- Understand how a tree is grown by an exhaustive search
- Know where and how the split point is set

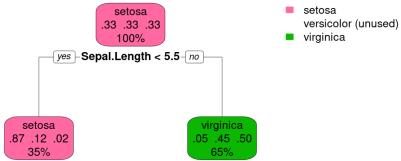
# TREE GROWING

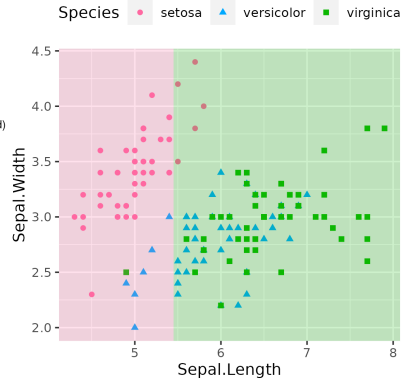
- We start with an empty tree, a root node that contains all the data. Trees are then grown by recursively applying **greedy** optimization to each node $\mathcal{N}$.
- Greedy means we do an **exhaustive search**: Ideally, all possible splits of $\mathcal{N}$ on all possible points $t$ for all features $x_j$ are compared in terms of their empirical risk $\mathcal{R}(\mathcal{N}, j, t)$.
- The training data is then distributed to child nodes according to the optimal split and the procedure is repeated in the child nodes.



# TREE GROWING

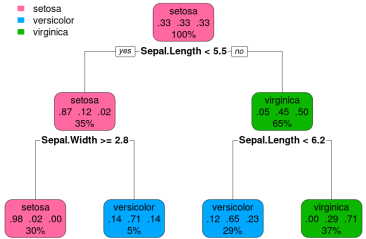
1. Start with a root node of all data.
2. Search for feature and split point that minimizes the empirical risk in child nodes – makes label distribution more homogenous.

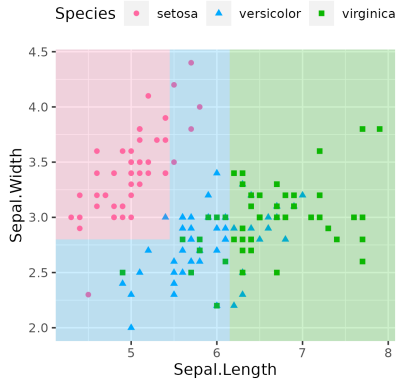


# TREE GROWING

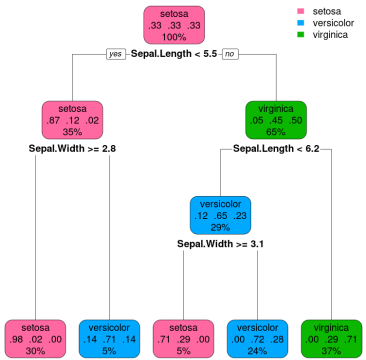
3. Proceed recursively for each child node: Select best split and divide data from parent node into left and right child nodes.

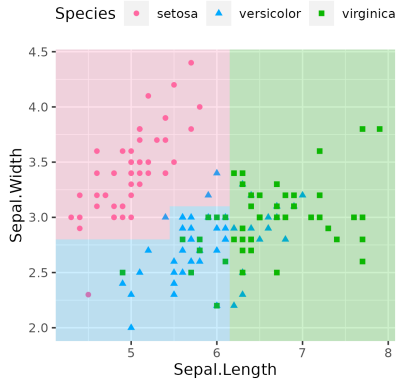


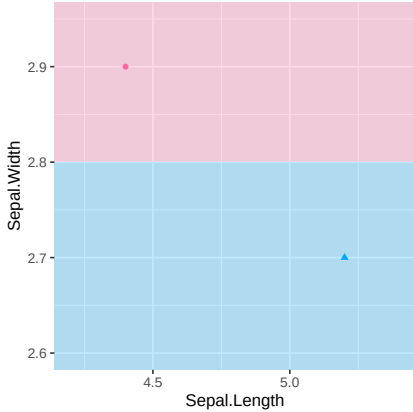
# TREE GROWING

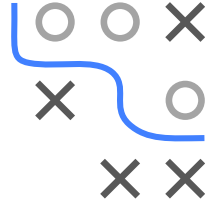
4. Repeat until we reach a stop criterion, e.g., until each leaf cannot be split further.



# SPLIT PLACEMENT

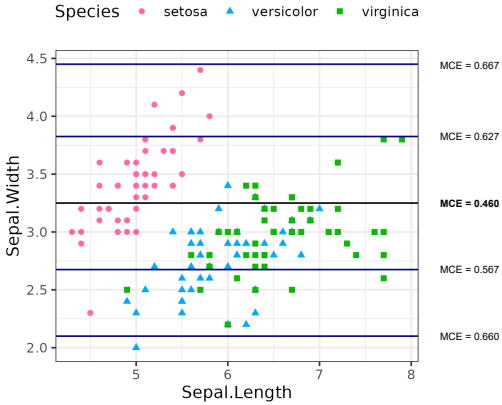


Splits are usually placed at the mid-point of the observations they split: the large margin to the next closest observations makes better generalization on new, unseen data more likely.

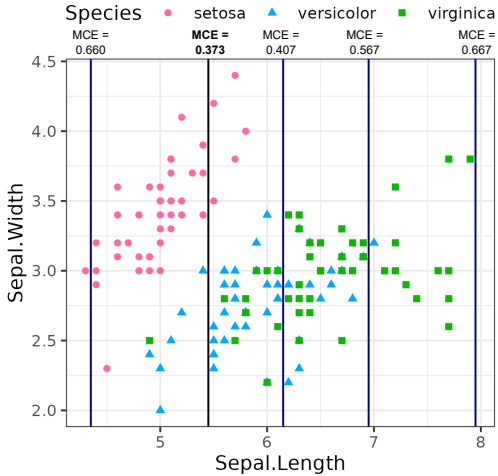
# FINDING THE SPLIT

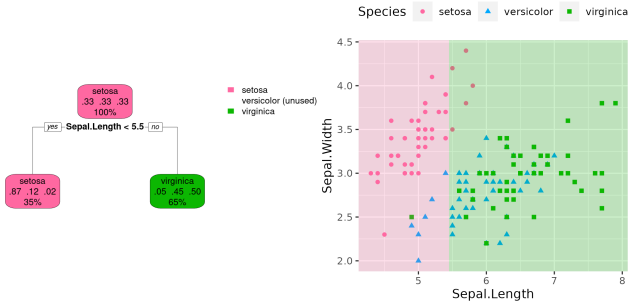
Assume we split the data so that the misclassification error (MCE) is minimal through the splitting.
First, we check a set of potential splits for `Sepal.Width`



# FINDING THE SPLIT

Then we check a set of potential splits for `Sepal.Length`



# FINDING THE SPLIT

- We take the split with lowest MCE: `Sepal.Length` = 5.5
- In real life, we actually search over many more splitting points. Common strategies involve: a) Searching over all possible split points (exhaustive search), b) searching quantile-wise
- MCE is rarely used, we will cover split criteria in detail later.